

Bangladesh Knowledge Base

RAG vs Base Model - Evidence and Analysis

VERDICT	PASS
RAG facts	24 / 24
Base facts	2 / 24
Required sources	5 / 5

Expected facts stated



So What?

RAG menjawab seluruh 24 fakta yang telah didefinisikan dan selalu mengambil dokumen yang diwajibkan. Base model hanya menyebut 2 dari 24 fakta dan menghasilkan beberapa detail yang terdengar meyakinkan tetapi bertentangan dengan dokumen. Untuk pertanyaan yang bergantung pada source-specific numbers, names, dan itinerary, retrieval mengubah jawaban dari plausible menjadi auditable.

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MAIN 2026 Phase 2 - AI Native Software Engineer Bootcamp

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1. Scope dan Corpus

Homework meminta minimal tiga dokumen tambahan, lima pertanyaan baru, dan perbandingan langsung antara jawaban RAG dan base model. Corpus expansion menggunakan tiga PDF resmi Bangladesh Tourism Board; delapan dokumen lama tetap dipertahankan agar perluasan basis pengetahuan dapat dibuktikan.

Document	Pages	Evaluation coverage	Integrity
Bangladesh Tourist Hand Book	76	Q1, Q2	SHA-256 verified
Sundarbans	8	Q3, Q4	SHA-256 verified
World Heritage Tour - 12 Nights, 13 Days	2	Q5	SHA-256 verified

Runtime proof

- 11 documents ingested: 8 pre-expansion documents plus 3 Bangladesh PDFs.
- 127 chunks stored; 127 of 127 chunks have Amazon Titan embeddings.
- Amazon Nova Lite generated both RAG and base-model answers in ap-southeast-2.
- Local vector-store RAG is verified live; the managed S3/Bedrock Knowledge Base path remains unverified because IAM credentials, bucket, Knowledge Base ID, and data source ID are not configured.

Evaluation contract

- Frozen dataset revision 2026-09-02-bangladesh-v2 with SHA-256 031cc8e6e2a01db877553db3ecbff9a55cd5903ab4d9bf16b4d73d0b0e2ffc9e.
- Same five inputs and same foundation model for both paths; RAG alone receives retrieved context.
- Temperature 0.0 and maximum 500 answer tokens.
- Deterministic graders check exact names, figures, source presence, and required-document match.
- Targeted, regression, and holdout splits are preserved; results are a five-case convenience sample, not a production failure-rate estimate.

2.1 Q1 - Case comparison

According to the Bangladesh Tourist Hand Book, how far is Khagrachari from Chattogram, which three rivers pass through the district, and what is its main attraction?

CASE	bangladesh.khagrachari.001	SPLIT	targeted
SOURCE	bangladesh-tourist-handbook.pdf	PDF PAGE	38

Expected evidence

- Khagrachari is 112 km from Chattogram
- Chengi River
- Kasalong River
- Maini River
- Alutila mysterious cave is the main attraction

RAG - 5/5 facts	Base model - 0/5 facts
Khagrachari is 112 km from Chattogram. The three rivers that pass through the district are Chengi, Kasalong, and Maini. The main attraction of Khagrachari is Alutila, a mysterious cave. [Source: bangladesh-tourist-handbook.pdf, PDF page 38]	Khagrachari is about 320 km from Chattogram. The three rivers that pass through the district are the Sangu, Karnaphuli, and Matamuhuri. The main attraction in Khagrachari is the Jhalong Hill Tracts, known for its scenic beauty and diverse tribal culture.

Measured result

RAG stated +5 more expected fact(s). Required source retrieved: yes.

Retrieved source: Bangladesh Tourist Hand Book, Bangladesh Tourism Board, PDF page 38. The required file appeared in the retrieval set.

2.2 Q2 - Case comparison

According to the Bangladesh Tourist Hand Book, what is Saint Martin's Island's national status and nickname, and what happens to Cheera-Dwip during tides?

CASE	bangladesh.saint-martin.002	SPLIT	targeted
SOURCE	bangladesh-tourist-handbook.pdf	PDF PAGE	34

Expected evidence

- Bangladesh's only coral island
- Nickname Pearl of Sea
- Cheera-Dwip is part of Saint Martin's Island
- Cheera-Dwip is divided during tides

RAG - 4/4 facts	Base model - 1/4 facts
Saint Martin's Island is the only coral island of Bangladesh and is known as the "Pearl of the Sea." Cheera-Dwip is a part of St. Martin's Island but is divided during tides. [Source: bangladesh-tourist-handbook.pdf], PDF page 34	According to the Bangladesh Tourist Hand Book, Saint Martin's Island is the only coral reef in Bangladesh and is often referred to as the "Maldives of Bangladesh." Its national status is that of a protected marine area. Cheera-Dwip, a part of the island, disappears during high tide and reappears during low tide, making it a unique tidal phenomenon.

Measured result

RAG stated +3 more expected fact(s). Required source retrieved: yes.

Retrieved source: Bangladesh Tourist Hand Book, Bangladesh Tourism Board, PDF page 34. The required file appeared in the retrieval set.

2.3 Q3 - Case comparison

According to the Sundarbans brochure, whose permission is needed before entering through Karamjal, what official station is Karamjal, and what kind of breeding center does it also serve as?

CASE	bangladesh.karamjal.003-v2	SPLIT	targeted
SOURCE	bangladesh-sundarbans.pdf	PDF PAGE	7

Expected evidence

- Permission from the local forest station
- Forest Rangers station
- Deer-breeding center

RAG - 3/3 facts	Base model - 0/3 facts
To enter the Sundarbans through Karamjal, permission from the local forest station is needed. Karamjal is a Forest Rangers station and also serves as a deer-breeding center. [Source: bangladesh-sundarbans.pdf]	To enter the Sundarbans through Karamjal, you need permission from the Forest Department. Karamjal is an official forest station and also serves as a crocodile breeding center. This dual role helps in both wildlife conservation and research.

Measured result

RAG stated +3 more expected fact(s). Required source retrieved: yes.

Retrieved source: Sundarbans, Bangladesh Tourism Board, PDF page 7. The required file appeared in the retrieval set.

2.4 Q4 - Case comparison

According to the Sundarbans brochure, which destination is about an 11-12 hour river journey from Khulna, what is its other name, and which forest range contains it?

CASE	bangladesh.kachikhali.004	SPLIT	regression
SOURCE	bangladesh-sundarbans.pdf	PDF PAGE	8

Expected evidence

- Kachikhali
- 11-12 hour river journey from Khulna
- Other name Tiger Point
- Sharonkhola Forest Range

RAG - 4/4 facts	Base model - 1/4 facts
Kachikhali is about an 11-12 hour river journey from Khulna. It is also known as Tiger Point and is located in the Sharonkhola Forest range. [Source: bangladesh-sundarbans.pdf]	The destination you're referring to is "Mongla," also known as "Mongla Port." It is part of the Sundarbans forest range. The Sundarbans is the world's largest mangrove forest, spanning parts of Bangladesh and India. The river journey from Khulna to Mongla takes about 11-12 hours, offering a unique way to experience the natural beauty and biodiversity of the Sundarbans.

Measured result

RAG stated +3 more expected fact(s). Required source retrieved: yes.

Retrieved source: Sundarbans, Bangladesh Tourism Board, PDF page 8. The required file appeared in the retrieval set.

2.5 Q5 - Case comparison

In the World Heritage Tour document, what travel time and distance are listed on day 5 for Khulna to Bagerhat and Bagerhat to Mongla, and which four Bagerhat heritage stops are named?

CASE	bangladesh.world-heritage-day-5.005	SPLIT	holdout
SOURCE	bangladesh-world-heritage-tour.pdf	PDF PAGE	1

Expected evidence

- 45 minutes from Khulna to Bagerhat
- 30 km from Khulna to Bagerhat
- 1 hour from Bagerhat to Mongla
- 52 km from Bagerhat to Mongla
- 60-domed Mosque
- Tomb of Khan Jahan Ali
- 9-domed Mosque
- Thakur's Pond

RAG - 8/8 facts	Base model - 0/8 facts
On Day 5, the travel time from Khulna to Bagerhat is 45 minutes, covering a distance of 30 km. The travel time from Bagerhat to Mongla is 1 hour, covering a distance of 52 km. The four Bagerhat heritage stops named are the 60-domed Mosque, Tomb of Khan Jahan Ali, 9-domed Mosque, and Thakur's Pond. [Source: bangladesh-world-heritage-tour.pdf]	On day 5, the travel time from Khulna to Bagerhat is approximately 2.5 hours, covering a distance of about 120 km. From Bagerhat to Mongla, the travel time is around 1.5 hours, covering about 60 km. The four heritage stops in Bagerhat are: Sixty Dome Mosque, Shat Gambuj Mosque, Khan Jahan Ali's Tomb, and Singar Mosque.

Measured result

RAG stated +8 more expected fact(s). Required source retrieved: yes.

Retrieved source: World Heritage Tour - 12 Nights, 13 Days, Bangladesh Tourism Board, PDF page 1. The required file appeared in the retrieval set.

3. Findings dan Decision

- Retrieval materially improved grounded completeness: 24/24 expected facts versus 2/24 for the base model.
- The base model produced confident but document-inconsistent details in all five cases, including wrong distances, river names, destination identity, breeding-center type, and itinerary figures.
- Required source coverage was 5/5. Passing facts without the assigned source would not have earned grounding credit.
- The final hybrid ranker combines semantic similarity with lexical overlap. This fixed exact-name and exact-number retrieval while retaining semantic matching for paraphrases.
- The holdout itinerary case remained 8/8 after targeted and regression improvements, so the candidate did not trade away the unseen case.

Verdict

PASS untuk scope homework local RAG: tiga dokumen Bangladesh berhasil ditambahkan, lima paired questions dijalankan, seluruh expected facts dan required sources terpenuhi, dan jawaban mentah dipertahankan dalam evidence. Managed S3/Bedrock Knowledge Base belum boleh disebut selesai sampai credential IAM dan tiga identifier AWS tersedia lalu ingestion mencapai COMPLETE.

Limitations

- The source PDFs are brochures and may be stale. This report evaluates faithfulness to the assigned documents, not present-day travel validity.
- The deterministic grader accepts declared string variants. It does not replace expert review for nuanced semantic quality.
- Five cases demonstrate the homework behavior but cannot estimate production reliability.
- No open-content licence was found on the source pages; files retain publisher attribution and are used for the educational assignment.

4. Reproduction dan Sources

Core commands

- cd backend
- ..\venv\Scripts\python.exe ingest_knowledge.py
- ..\venv\Scripts\python.exe compare_rag_vs_base.py
- cd ..
- ..\venv\Scripts\python.exe -m unittest discover -s tests -v

Official source files

Bangladesh Tourist Hand Book (76 pages)

Publisher: Bangladesh Tourism Board

SHA-256: F7F94802736536F6F34806A47E9AF58607F03B45C1836A25286B94D1484DDF6F

[Open official PDF](#)

Sundarbans (8 pages)

Publisher: Bangladesh Tourism Board

SHA-256: 82FFB983FBCE26734027E80977F1B056C3BD21C765FF784CC1D8D44B2001433A

[Open official PDF](#)

World Heritage Tour - 12 Nights, 13 Days (2 pages)

Publisher: Bangladesh Tourism Board

SHA-256: BD79FA6C257D6224E36BAFD124B2A3E00A730E5A21D81D1CBC2D48EBD0D72184

[Open official PDF](#)

Raw evidence: evidence/session-09-rag-vs-base.md | Run at 2026-09-02 05:39 UTC | Retrieval mode local-vector-store